



Modelling Financial Markets

Seminar 7: Value-at-Risk and Expected Shortfall — Hands-on



Daniel Traian PELE

Drd. Antoaneta Amza, antoaneta.amza@csie.ase.ro

Bucharest University of Economic Studies

IDA Institute Digital Assets

Blockchain Research Center

AI4EFin Artificial Intelligence for Energy Finance

Romanian Academy, Institute for Economic Forecasting

MSCA Digital Finance

Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ BVB: Bursa de Valori București (the Bucharest Stock Exchange)
- ▣ CF: Cornish–Fisher (expansion)
- ▣ CI: Confidence Interval
- ▣ COVID-19: COronaVirus Disease 2019
- ▣ CV: Cross-Validation
- ▣ DM: Diebold–Mariano (test)
- ▣ ES: Expected Shortfall
- ▣ EVT: Extreme Value Theory
- ▣ FHS: Filtered Historical Simulation
- ▣ FRTB: Fundamental Review of the Trading Book
- ▣ FZ0: Fissler–Ziegel loss, zero-homogeneous version
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GPD: Generalised Pareto Distribution
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ LGD: Loss Given Default
- ▣ OMV: Österreichische Mineralölverwaltung (Austrian Mineral Oil Administration (OMV group))
- ▣ QML: Quasi-Maximum Likelihood
- ▣ RMSFE: Root Mean Squared Forecast Error
- ▣ SAV: Symmetric Absolute Value (CAViaR specification)
- ▣ SE: Standard Error
- ▣ XFG: Quantlet prefix of Applied Quantitative Finance (Härdle, Chen & Overbeck)

Outline

Introduction

Part A: Derivations

Part B: Estimation and Inference

Part C: Open Questions

Wrap-up

References

Seminar Objectives

By the end of this seminar you can

- ▣ Derive VaR and ES for the Normal and Student- t distributions and the sampling variance of their historical estimates
- ▣ Prove subadditivity of ES from the Rockafellar–Uryasev representation
- ▣ Estimate historical, filtered, extreme-value and CAViaR VaR/ES with honest intervals (HAC, block and residual bootstrap, declustering)
- ▣ Split portfolio risk into Euler contributions

- ▣ Background: Lecture 7; textbooks McNeil, Frey & Embrechts (2015) and Franke, Härdle & Hafner (2019)

Structure

- ▣ **Part A:** derivations (A1–A8)
- ▣ **Part B:** seven computational exercises with inference (B1–B7)
- ▣ **Part C:** two open questions, seeds for the team project (C1, C3), and a critique of an AI answer (C2)

How to Work in This Seminar

- **[Solved]** problems show the full worked solution; **[Proposed]** problems show only the task: their solutions are discussed in class
- Every proposed problem names its **model**: the solved problem that uses the same technique
 - ▶ Every Part B exercise ends with interpretation questions
- Sign convention: loss $L = -r$ in % of the position; VaR and ES are positive numbers
- Report every tail estimate with its uncertainty: how many observations lie beyond it?

Solved and Proposed Problems

Solved: worked examples in this deck

- Part A: A1 (Normal ES derivation, SE of historical VaR), A2 (Student- t ES derivation), A3 (a discrete loss), A5 (component VaR; (d) ES contributions)
- Part B: B1 (historical VaR and ES on BET; bootstrap, asymptotic and HAC intervals), B2 (GPD fit on BET; extremal index)

Proposed: you solve them

- Part A: A4 (failure of subadditivity; model: A3), A6 (validity of Cornish–Fisher; model: A1), A7 (ES as a minimum; model: A3, A4), A8 (variance of historical ES; model: A1)
- Part B: B3 (filtered vs historical on Bitcoin; model: B1), B4 (Cornish–Fisher vs empirical; model: A1, B1), B5 (component VaR of a portfolio; model: A5), B6 (bootstrap interval for ES; model: B1), B7 (CAViaR on BET; model: B3)
- Part C: C1, open question and project seed; C2, critique of an AI answer (model: B1); C3, deep quantile regression for VaR, project seed from the lecture case study

Setup: Python Environment and Data

-  **Open in Colab:** `chapter7_seminar_notebook.ipynb`
- Google Colab (recommended) or local Python ≥ 3.10 with `numpy`, `pandas`, `scipy`, `arch`, `matplotlib`
- Series: BET (2000–2026), S&P 500, Bitcoin (2014–2026), EUR/RON (BNR reference rate), gold; SPY, TLT, GLD; eight BVB blue chips; daily closes up to 18 September 2026
- Equity indices: weekdays only; holiday records (unchanged close, zero volume) removed, genuine unchanged closes kept as zero returns; crypto keeps all 7 days of the week
- Portfolios: prices joined on common trading days first, then simple returns

Formulas Used Today

- Level α = tail probability (VaR 1%, ES 2.5%); return X , loss $L = -X$: $\text{VaR}_\alpha(X) = -\inf\{x : P(X \leq x) > \alpha\} = -q_\alpha(X)$,
 $\text{ES}_\alpha(X) = -\frac{1}{\alpha} \int_0^\alpha q_u(X) du$
- Discrete case [Acerbi & Tasche \(2002\)](#): $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L1\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha(\alpha - P(L > \text{VaR}_\alpha)))$
- Normal distribution, $z_\alpha = \Phi^{-1}(\alpha) < 0$: $\text{VaR}_\alpha = -(\mu + \sigma z_\alpha)$, $\text{ES}_\alpha = -\mu + \sigma \varphi(z_\alpha)/\alpha$
- Student- t : $\text{VaR}_\alpha = -(m + s t_\alpha)$, $\text{ES}_\alpha = -m + s \frac{\varphi_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$; variance σ^2 when $s = \sigma \sqrt{(\nu - 2)/\nu}$
- GPD tail of the losses: $\text{VaR}_\alpha = u + \frac{\beta}{\xi} \left[\left(\frac{n}{N_u} \alpha \right)^{-\xi} - 1 \right]$, $\text{ES}_\alpha = \frac{\text{VaR}_{\alpha+\beta-\xi u}}{1-\xi}$ ($\alpha < N_u/n$; ES needs $\xi < 1$)
- Euler: $C_i = w_i \partial \text{VaR} / \partial w_i = \mathbf{z}_{1-\alpha} w_i (\Sigma w)_i / \sigma_p$, $\sum_i C_i = \text{VaR}$ [Tasche \(2008\)](#)

A1: Normal VaR and ES [Solved]: Task

□ Data and assumptions

- ▶ Daily returns of a position are Normal, $X \sim N(\mu, \sigma^2)$, with $\mu = 0.04\%$ and $\sigma = 1.2\%$; the loss is $L = -X$.
- ▶ We write $z_\alpha = \Phi^{-1}(\alpha) < 0$, and Φ , φ for the standard Normal distribution and density functions.

□ Tasks

- ▶ (a) Derive $ES_\alpha = -\mu + \sigma\varphi(z_\alpha)/\alpha$ starting from the definition $ES_\alpha = -\mathbb{E}[X \mid X \leq q_\alpha(X)]$.
- ▶ (b) Compute VaR and ES at $\alpha = 1\%$ and at $\alpha = 2.5\%$, in % of the position.
- ▶ (c) Compute the asymptotic standard error of the *historical* VaR 1%, $SE = \sqrt{\alpha(1-\alpha)/n}/f(\text{VaR}_\alpha)$, for $n = 500$ and for $n = 6\,670$ days, and the width of the 95% interval.

□ Report

- ▶ The four numbers of (b), the two standard errors of (c), and a comparison with the bootstrap interval widths of B1: why are the B1 intervals wider?

A1: Normal VaR and ES [Solved]: Solution (1/2)

- **(a) Derivation.** Write $X = \mu + \sigma Z$; the event $X \leq q_\alpha$ is $Z \leq z_\alpha$, which has probability α .
 - ▶ Since $\varphi'(u) = -u\varphi(u)$, we get $\int_{-\infty}^z u\varphi(u) du = [-\varphi(u)]_{-\infty}^z = -\varphi(z)$.
 - ▶ Hence $\mathbb{E}[X \mid X \leq q_\alpha] = \mu + \frac{\sigma}{\alpha} \int_{-\infty}^{z_\alpha} u\varphi(u) du = \mu - \sigma \frac{\varphi(z_\alpha)}{\alpha}$, and changing the sign gives the formula.
- **(b) Numbers at 1%.** The quantile is $z_{1\%} = -2.326$ and the density there is $\varphi(z_{1\%}) = 0.0267$, so $\varphi/\alpha = 2.665$.
 - ▶ $\text{VaR}_{1\%} = -(0.04 + 1.2 \cdot (-2.326)) = 2.75\%$ and $\text{ES}_{1\%} = -0.04 + 1.2 \cdot 2.665 = 3.16\%$.
- **(b) Numbers at 2.5%.** Here $z_{2.5\%} = -1.960$ and $\varphi/\alpha = 2.338$.
 - ▶ $\text{VaR}_{2.5\%} = 2.31\%$ and $\text{ES}_{2.5\%} = -0.04 + 1.2 \cdot 2.338 = 2.77\%$.
 - ▶ Interpretation: ES 2.5% (2.77%) is almost equal to VaR 1% (2.75%); this near-equality under the Normal distribution is why FRTB chose ES 2.5%.

A1: Normal VaR and ES [Solved]: Solution (2/2)

- **(c) Density at the quantile.** The loss density at VaR 1% is $f(\text{VaR}_{1\%}) = \varphi(z_{1\%})/\sigma = 0.0267/1.2 = 0.0222$.
 - ▶ $n = 500$: $\text{SE} = \sqrt{0.01 \cdot 0.99/500}/0.0222 = 0.20$ percentage points, so the 95% interval is 0.79 points wide.
 - ▶ $n = 6\,670$: $\text{SE} = 0.055$ and the 95% width is 0.22 points.
- **Comparison with B1.** The bootstrap intervals for BET VaR 1% are 1.01 points wide (500 days) and 0.61 points wide (full sample).
 - ▶ Interpretation: BET losses have a heavier tail than the Normal distribution, so the density at VaR is lower and the quantile is estimated less precisely.
 - ▶ The standard error falls like $1/\sqrt{n}$: thirteen times more data make the interval about $\sqrt{13.3} \approx 3.7$ times narrower.

A2: Student- t ES [Solved]: Task

□ Data and assumptions

- ▶ Returns are $X = m + sT$ with T Student- t with ν degrees of freedom and density $g_\nu(x) = c_\nu(1 + x^2/\nu)^{-(\nu+1)/2}$.
- ▶ The quantile is $t_\alpha = t_\nu^{-1}(\alpha) < 0$; for the numbers use $\nu = 4$, $m = 0.04\%$ and the same standard deviation 1.2% as in A1.

□ Tasks

- ▶ (a) Show that $\int_{-\infty}^t x g_\nu(x) dx = -g_\nu(t) \frac{\nu + t^2}{\nu - 1}$ for $\nu > 1$, and deduce $\text{ES}_\alpha = -m + s \frac{g_\nu(t_\alpha)}{\alpha} \frac{\nu + t_\alpha^2}{\nu - 1}$.
- ▶ (b) Find the scale s that gives a standard deviation of 1.2%, using $\text{Var}(T) = \nu/(\nu - 2)$.
- ▶ (c) Compute VaR and ES at $\alpha = 1\%$ and $\alpha = 2.5\%$.

□ Report

- ▶ The four numbers of (c) next to those of A1; interpretation question: at which level do the two models differ most, and why?

A2: Student- t ES [Solved]: Solution

- (a) **Antiderivative.** Differentiate $(1 + x^2/\nu)^{-(\nu-1)/2}$: the result is $-\frac{\nu-1}{\nu} x (1 + x^2/\nu)^{-(\nu+1)/2}$.
 - Therefore $xg_\nu(x) = -\frac{\nu c_\nu}{\nu-1} \frac{d}{dx} (1 + x^2/\nu)^{-(\nu-1)/2}$, and integrating from $-\infty$ to t gives $-\frac{\nu}{\nu-1} g_\nu(t)(1 + t^2/\nu)$.
 - This equals $-g_\nu(t)(\nu + t^2)/(\nu-1)$; the limit at $-\infty$ vanishes only if $\nu > 1$. Divide by α and rescale by $m + sT$ as in A1.
- (b) **Scale.** $s^2\nu/(\nu-2) = 1.2^2$, so $s = 1.2\sqrt{2/4} = 0.849$.
- (c) **At 1%.** $t_{1\%} = -3.747$ and $g_4(t_{1\%}) = 0.0087$, so the ES factor is $\frac{0.0087}{0.01} \cdot \frac{4 + 14.04}{3} = 5.221$.
 - $\text{VaR}_{1\%} = -(0.04 + 0.849 \cdot (-3.747)) = 3.14\%$ and $\text{ES}_{1\%} = -0.04 + 0.849 \cdot 5.221 = 4.39\%$.
- (c) **At 2.5%.** $t_{2.5\%} = -2.776$ gives $\text{VaR}_{2.5\%} = 2.32\%$ and $\text{ES}_{2.5\%} = 3.35\%$.
 - Interpretation: VaR 2.5% is almost the same in both models (2.32 vs 2.31), while VaR 1% and both ES values are much higher under t_4 : heavy tails show only deep in the tail, and ES sees them first.

A3: VaR and ES of a Discrete Loss [Solved]: Task

□ Data and assumptions

- ▶ A bond loses 60 (loss given default, LGD) with probability $p = 3\%$ and 0 otherwise, so the return is $X = -60$ or $X = 0$.
- ▶ Use the general definitions $\text{VaR}_\alpha = -\inf\{x : P(X \leq x) > \alpha\}$ and the discrete ES formula of the lecture [Acerbi & Tasche \(2002\)](#).

□ Tasks

- ▶ (a) Compute VaR 5% and VaR 2.5% directly from the definition.
- ▶ (b) Compute ES 5% and ES 2.5% with $\text{ES}_\alpha = \frac{1}{\alpha} (\mathbb{E}[L1\{L > \text{VaR}_\alpha\}] + \text{VaR}_\alpha(\alpha - P(L > \text{VaR}_\alpha)))$.
- ▶ (c) Explain why $\text{ES}_{5\%}$ is not equal to $\mathbb{E}[L \mid L \geq \text{VaR}_{5\%}]$.

□ Report

- ▶ The four numbers and a one-sentence answer to (c).

A3: VaR and ES of a Discrete Loss [Solved]: Solution

- (a) **VaR 5%.** $P(X \leq -60) = 3\%$ is not above 5%, while $P(X \leq 0) = 100\% > 5\%$; so $q_{5\%} = 0$ and $\text{VaR}_{5\%} = 0$.
 - ▶ VaR 2.5%: $P(X \leq -60) = 3\% > 2.5\%$, so $q_{2.5\%} = -60$ and $\text{VaR}_{2.5\%} = 60$.
- (b) **ES 5%.** $\mathbb{E}[L1\{L > 0\}] = 60 \cdot 0.03 = 1.8$ and $P(L > 0) = 3\%$, so $\text{ES}_{5\%} = \frac{1.8 + 0 \cdot (0.05 - 0.03)}{0.05} = 36$.
 - ▶ ES 2.5%: nothing lies above 60, so $\text{ES}_{2.5\%} = \frac{0 + 60 \cdot (0.025 - 0)}{0.025} = 60$.
- (c) **Why the correction term.** $\mathbb{E}[L \mid L \geq 0] = 1.8$ averages over **all** days, not over the worst 5%.
 - ▶ Interpretation: with atoms, the worst 5% of outcomes consist of the 3% default mass plus 2% of the zero-loss mass; the formula weights them exactly, and ES then equals 36.

A4: When VaR Punishes Diversification [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ Two independent bonds A and B; each loses 100 with probability $p = 0.7\%$ and 0 otherwise [Artzner et al. \(1999\)](#). The portfolio is $A + B$.

□ Tasks

- ▶ (a) Write the distribution of the portfolio loss (values 0, 100, 200 and their probabilities).
- ▶ (b) Compute VaR 1% and ES 1% of one bond and of the portfolio, with the discrete formula of A3.
- ▶ (c) Show that VaR 1% is not subadditive here while ES 1% is.
- ▶ (d) Repeat (b)–(c) at $\alpha = 1.5\%$ and find the range of α for which VaR fails.

□ Report

- ▶ A small table of VaR and ES for A, B and $A + B$ at both levels, and one sentence on what a VaR-based limit would reward.
- ▶ Model: A3.

A5: Component VaR and ES of Two Assets [Solved]: Task

□ Data and assumptions

- ▶ Equity 600 000 EUR with $\sigma_1 = 1.2\%$ daily; bonds 400 000 EUR with $\sigma_2 = 0.9\%$; correlation $\rho = -0.2$; Normal distribution, zero mean, $\alpha = 1\%$.

□ Tasks

- ▶ (a) Compute the portfolio standard deviation in EUR and the portfolio VaR 1%.
- ▶ (b) Compute the marginal VaR $z_{1-\alpha}(\Sigma w)_i/\sigma_p$ and the component VaR C_i of each asset, and check $C_1 + C_2 = \text{VaR}$ Tasche (2008).
- ▶ (c) Compute the diversification benefit (sum of stand-alone VaRs minus portfolio VaR).
- ▶ (d) Using $\text{ES}_\alpha(w) = \min_v \{v + \mathbb{E}[(L_p - v)_+]/\alpha\}$ Rockafellar & Uryasev (2000), show that $w_i \partial \text{ES}_\alpha / \partial w_i = \mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_\alpha]$ and evaluate these ES contributions here.

□ Report

- ▶ A table of C_i and C_i^{ES} with shares; interpretation: do the bonds carry risk in proportion to their money weight?

A5: Component VaR and ES of Two Assets [Solved]: Solution (1/2)

- (a) **Portfolio risk.** Stand-alone standard deviations: $600\,000 \cdot 1.2\% = 7\,200$ EUR and $400\,000 \cdot 0.9\% = 3\,600$ EUR.
 - ▶ $\sigma_p^2 = 7\,200^2 + 3\,600^2 + 2(-0.2)(7\,200)(3\,600) = 54\,432\,000$, so $\sigma_p = 7\,378$ EUR and $\text{VaR}_{1\%} = 2.326 \cdot 7\,378 = 17\,163$ EUR.
- (b) **Marginal and component VaR.** $(\Sigma w) = (77.76, 19.44)$ (in EUR, with Σ in decimal units), which gives the marginal VaR per EUR of exposure $(0.0245, 0.0061)$.
 - ▶ $C_1 = 14\,711$ EUR (85.7%) and $C_2 = 2\,452$ EUR (14.3%); their sum is 17 163 EUR, exactly the portfolio VaR.
- (c) **Diversification.** Stand-alone VaRs $16\,750 + 8\,375 = 25\,125$ EUR against a portfolio VaR of 17 163 EUR.
 - ▶ Interpretation: the bonds hold 40% of the money but only 14.3% of the risk, because the negative correlation makes them a partial hedge.

A5: Component VaR and ES of Two Assets [Solved]: Solution (2/2)

- **(d) Envelope argument.** $v^* = \text{VaR}_\alpha$ minimises $G(w, v) = v + \mathbb{E}[(L_p - v)_+]/\alpha$, so $\partial \text{ES}_\alpha / \partial w_i = \partial G / \partial w_i$ at v^* .
 - ▶ Differentiating under the expectation: $\partial G / \partial w_i = \mathbb{E}[-R_i \mathbf{1}\{L_p > v^*\}]/\alpha$.
 - ▶ For a continuous L_p , $P(L_p > \text{VaR}_\alpha) = \alpha$, so this is $\mathbb{E}[-R_i \mid L_p \geq \text{VaR}_\alpha]$; multiplying by w_i and summing gives $\mathbb{E}[L_p \mid L_p \geq \text{VaR}_\alpha] = \text{ES}_\alpha$.
- **(d) Numbers.** Under the Normal distribution $\text{ES}_\alpha = \sigma_p \varphi(z_\alpha) / \alpha$, so every contribution scales by $\frac{\varphi(z_\alpha) / \alpha}{z_{1-\alpha}} = 1.146$.
 - ▶ $C_1^{\text{ES}} = 16\,854$ EUR and $C_2^{\text{ES}} = 2\,809$ EUR; together 19 663 EUR, the portfolio ES 1%.
 - ▶ Interpretation: under the Normal distribution the ES shares equal the VaR shares, so ES adds no new allocation information; with heavy tails it does (see B5).

A6: When Is Cornish–Fisher a Valid Quantile? [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ Returns with $\mu = 0$, $\sigma = 1.2\%$, skewness $S = -0.5$ and excess kurtosis K [Cornish & Fisher \(1938\)](#).
- ▶ The Cornish–Fisher map is $\tilde{z}(z) = z + \frac{(z^2 - 1)S}{6} + \frac{(z^3 - 3z)K}{24} - \frac{(2z^3 - 5z)S^2}{36}$ and $\text{VaR}_\alpha = -\sigma\tilde{z}(z_\alpha)$.

□ Tasks

- ▶ (a) With $K = 3$, compute the three correction terms at $z_{1\%}$, then $\tilde{z}_{1\%}$ and the Cornish–Fisher VaR 1%; compare with the Normal VaR.
- ▶ (b) Derive $d\tilde{z}/dz$ and find where it is smallest on $[z_{0.1\%}, 0]$.
- ▶ (c) Find the largest K for which $\tilde{z}$ is increasing on this interval, and decide whether $K = 10$ (close to the S&P 500) is admissible.

□ Report

- ▶ The VaR of (a), the bound on K , and whether a VaR computed with $K = 10$ is a valid estimate. Model: A1.

A7: ES as a Minimum, and Subadditivity [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ L is any integrable loss; define $F_L(v) = v + \mathbb{E}[(L - v)_+]/\alpha$ Rockafellar & Uryasev (2000). For the numbers, use the A4 portfolio (two bonds, $p = 0.7\%$) at $\alpha = 1\%$.

□ Tasks

- ▶ (a) Prove that F_L is convex and that its minimum over v equals $\text{ES}_\alpha(L)$, attained at $v = \text{VaR}_\alpha(L)$.
- ▶ (b) Deduce $\text{ES}_\alpha(L_1 + L_2) \leq \text{ES}_\alpha(L_1) + \text{ES}_\alpha(L_2)$ from $(a + b)_+ \leq a_+ + b_+$.
- ▶ (c) For the A4 portfolio, compute the slope of F on $(0, 100)$ and on $(100, 200)$, the minimiser and the minimum.

□ Report

- ▶ The two proofs in a few lines and the three numbers of (c), compared with $\text{ES}_{1\%}(A + B)$ from A4. Model: A3, A4.

A8: The Sampling Variance of Historical ES [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ n i.i.d. losses with a continuous distribution, a positive density at VaR_α and $\mathbb{E}[L^2] < \infty$; historical ES is the mean of the losses beyond the empirical VaR. Its influence function is $\text{IF}(x) = \text{VaR}_\alpha + (x - \text{VaR}_\alpha)_+ / \alpha - \text{ES}_\alpha$ [Chen \(2008\)](#).

□ Tasks

- ▶ (a) Show that the variance of the limit of $\sqrt{n}(\widehat{\text{ES}}_\alpha - \text{ES}_\alpha)$ is $\text{avar}(\widehat{\text{ES}}_\alpha) = \text{Var}(\text{IF}(L)) = \text{Var}((L - \text{VaR}_\alpha)_+) / \alpha^2$, and explain why estimating VaR adds no term.
- ▶ (b) For $L \sim N(0, 1)$ and $\alpha = 2.5\%$, compute $\mathbb{E}[(L - v)_+]$ and $\mathbb{E}[(L - v)_+^2]$ in closed form at $v = \text{VaR}_\alpha$.
- ▶ (c) Compute the standard error of $\widehat{\text{ES}}_{2.5\%}$ for $n = 500$, for $\sigma = 1$ and for $\sigma = 1.2\%$, and compare with the S&P 500 over its last 500 days.

□ Report

- ▶ The two moments, the asymptotic variance and the two standard errors; interpretation: why is the S&P 500 value much larger? Model: A1.

B1: Historical VaR and ES on BET [Solved]: Task

□ Data and assumptions

- ▶ BET index (Bucharest Stock Exchange), data/market/BET.csv, daily closes 6 January 2000 – 18 September 2026; daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$, loss $L = -r$; weekdays only (the file has no holiday records).
- ▶ Two samples: the full period ($n = 6\,670$) and the last 500 days (from 2024-09-12).

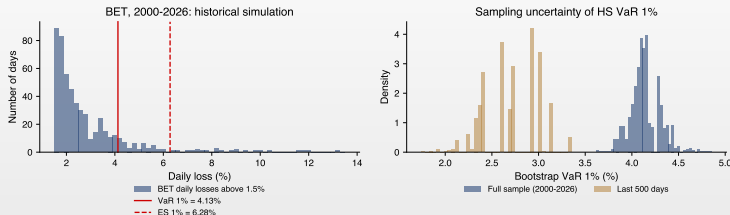
□ Tasks

- ▶ (a) Compute historical VaR and ES at $\alpha = 1\%$ and $\alpha = 2.5\%$ on both samples (VaR = empirical $(1 - \alpha)$ quantile of L , ES = mean of the losses at or above it).
- ▶ (b) Build a 95% percentile bootstrap interval for VaR 1% with 2000 i.i.d. resamples [Efron \(1979\)](#).
- ▶ (c) Build three more 95% intervals for VaR 1%: asymptotic i.i.d. ($\widehat{\text{VaR}} \pm 1.96 \sqrt{\alpha(1 - \alpha)/n/\hat{f}}$, Gaussian kernel density, Silverman bandwidth), asymptotic HAC (Bartlett kernel, lag $\lfloor 4(n/100)^{2/9} \rfloor$), and a moving-block bootstrap with 20-day blocks [Künsch \(1989\)](#).

□ Report

- ▶ A table with the eight VaR/ES numbers and the four intervals. Interpretation: how many losses lie beyond VaR 1% in each sample, which estimate would you report to a risk committee, and which interval is widest and why?

B1: Historical VaR and ES on BET [Solved]: Solution (1/3)



	n	$\text{VaR}_{1\%}$	$\text{ES}_{1\%}$	$\text{VaR}_{2.5\%}$	$\text{ES}_{2.5\%}$
2000–2026	6 670	4.13	6.28	2.80	4.57
Last 500 days	500	2.71	3.22	2.05	2.69

- **(a) Procedure.** Sort the losses, take the 99% (97.5%) empirical quantile as VaR and average the losses at or above it for ES; the chart shows the tail and the bootstrap distributions of VaR 1%.

B1: Historical VaR and ES on BET [Solved]: Solution (2/3)

- (a) **Tail counts.** Beyond VaR 1% lie 67 losses in the full sample but only 5 in the last 500 days.
 - ▶ So the 500-day ES 1% is the average of only five numbers and should be reported with a warning.
- (b) **Bootstrap intervals.** Full sample [3.85, 4.46]; last 500 days [2.11, 3.12].
 - ▶ The two intervals do not overlap: the recent market is calmer than the 26-year average.
 - ▶ Interpretation: report the recent value for today's risk and the long-run value as a stress reference.

B1: Historical VaR and ES on BET [Solved]: Solution (3/3)

	i.i.d. bootstrap	asympt., i.i.d.	asympt., HAC	block bootstrap
2000–2026	[3.85, 4.46]	[3.81, 4.44]	[3.69, 4.57]	[3.61, 4.80]
Last 500 days	[2.11, 3.12]	[2.18, 3.25]	[2.19, 3.23]	[2.02, 3.03]

- **(c) Asymptotic step.** $\hat{f}(\widehat{\text{VaR}}) = 0.0076$ (full) and 0.0163 (500 days), so $\text{SE} = \sqrt{0.01 \cdot 0.99/n/\hat{f}} = 0.16$ and 0.27 ; the HAC standard errors are 0.23 and 0.27 .
- **Reading the table.** Full-sample widths: $0.61, 0.63, 0.88, 1.19$; exceedances cluster in 2008 and 2020, so HAC and blocks widen the interval by a half to a factor of two.
- Last 500 days: all four widths are about 1; the period is calm, with little clustering, so the i.i.d. answer is adequate there.

B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Task

□ Data and assumptions

- ▶ BET index (Bucharest Stock Exchange), data/market/BET.csv, daily closes 6 January 2000 – 18 September 2026; daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$, loss $L = -r$; weekdays only (the file has no holiday records).
- ▶ Peaks over threshold: the excesses $L - u$ above a high threshold u follow a generalised Pareto distribution (GPD) with shape ξ and scale β [McNeil, Frey & Embrechts \(2015\)](#).

□ Tasks

- ▶ (a) Set u so that the largest 5% of losses lie above it and fit the GPD to the excesses by maximum likelihood.
- ▶ (b) Compute the i.i.d. standard errors $SE(\hat{\xi}) = (1 + \hat{\xi})/\sqrt{N_u}$ and $SE(\hat{\beta}) = \hat{\beta}\sqrt{2(1 + \hat{\xi})/N_u}$.
- ▶ (c) Compute the GPD VaR and ES at $\alpha = 1\%$, 0.5% and 0.1% and compare with historical simulation.
- ▶ (d) Refit for thresholds leaving from 10% down to 1.5% of the losses above them and plot $\hat{\xi}$ with 95% bands.
- ▶ (e) Estimate the extremal index θ of the exceedances of u with the intervals estimator [Ferro & Segers \(2003\)](#), decluster, and refit the GPD on the cluster maxima.

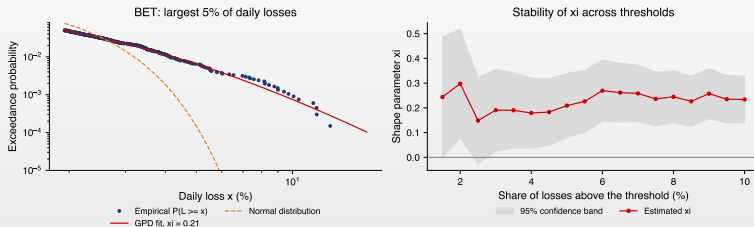
□ Report

- ▶ The fit, the VaR/ES table, the stability chart and the declustered fit. Interpretation: is the BET tail heavier than the Normal distribution, how sure are we, and is the i.i.d. SE of $\hat{\xi}$ credible?

B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Solution (1/3)

- (a) **Fit.** $u = 1.93\%$ leaves $N_u = 334$ of $n = 6\,670$ losses above it; maximum likelihood gives $\hat{\xi} = 0.209$ and $\hat{\beta} = 1.20$.
- (b) **Standard errors.** $SE(\hat{\xi}) = (1 + 0.209)/\sqrt{334} = 0.066$ and $SE(\hat{\beta}) = 0.10$.
 - ▶ $\hat{\xi}/SE = 3.2$: under independence the tail is significantly heavier than exponential (Normal-type); (e) checks the independence assumption.
- (c) **VaR 1%, step by step.** $N_u/n = 0.050$, so $(0.01/0.050)^{-0.209} = 1.401$.
 - ▶ $VaR_{1\%} = 1.93 + \frac{1.20}{0.209}(1.401 - 1) = 4.22\%$ (historical 4.13%); $ES_{1\%} = 6.34\%$ (historical 6.28%).
 - ▶ Deeper: VaR 0.5% 5.47 vs 5.38; VaR 0.1% 9.18 vs 9.50, with only 7 observations beyond; ES 0.1% 12.61 vs 11.25.
 - ▶ Interpretation: the two methods agree down to 0.5%; at 0.1% historical ES averages seven numbers, and only the GPD gives a stable value.

B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Solution (2/3)



- (d) **Stability.** Across thresholds $\hat{\xi}$ ranges from 0.15 to 0.30, but VaR 0.1% only from 9.04% to 9.51%.
- Reading the chart: left, the GPD follows the empirical tail and the Normal tail falls far below it; right, the band widens as fewer losses remain. The quantile is more stable than the shape parameter.

B2: GPD Fit and Extreme-Value VaR on BET [Solved]: Solution (3/3)

- (e) **Extremal index.** From the gaps T_i between the 334 exceedances: $\hat{\theta} = \frac{2(\sum_i (T_i - 1))^2}{(N - 1) \sum_i (T_i - 1)(T_i - 2)} = 0.32$.
- ▶ Declustering [Ferro & Segers \(2003\)](#): target $C = \lfloor \hat{\theta}N \rfloor + 1 = 108$ clusters, cut at the $C - 1$ largest gaps; the $(C - 1)$ -th and C -th largest gaps are equal, so C is reduced until they differ, leaving 100 clusters of 3.3 exceedances on average.
 - ▶ At the VaR 1% threshold $\hat{\theta} = 0.40$.
- **Declustered fit.** GPD on the 100 cluster maxima: $\hat{\xi} = 0.269$ (SE 0.127) against 0.209 (SE 0.066) on all exceedances.
- ▶ $\hat{\xi}/\text{SE}$ falls from 3.2 to 2.1: only about 100 of the 334 excesses carry independent information.
 - ▶ Interpretation: the tail is still heavy, but the i.i.d. SE of 0.066 is not credible; report the declustered fit, or filter with GARCH first (conditional EVT).

B3: Filtered vs Historical Simulation on Bitcoin [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ Bitcoin in USD, data/market/BTC-USD.CC.csv, daily closes 17 September 2014 – 18 September 2026, 7 days a week; daily log returns in %.
- ▶ Model: $r_t = \mu_t + \sigma_t z_t$ with AR(1) mean and GARCH(1,1) variance, estimated by quasi-maximum likelihood Barone-Adesi et al. (1999).

□ Tasks

- ▶ (a) Estimate the model on the full sample and forecast $\hat{\mu}_{t+1}$ and $\hat{\sigma}_{t+1}$ for 19 September 2026.
- ▶ (b) Compute the filtered VaR 1% and ES 1% from the empirical quantile of the standardised residuals, and compare with historical simulation on 500 days and on the full sample.
- ▶ (c) Using one-day-ahead forecasts (GARCH re-estimated each January), count the days of the last two years with a loss above each VaR.
- ▶ (d) Build a 90% residual-bootstrap interval for today's filtered VaR 1% with 999 paths, re-estimating the GARCH model on each path Christoffersen & Gonçalves (2005).

□ Report

- ▶ The three VaR/ES pairs, the exceedance counts and the interval. Interpretation: why can the three VaRs differ by a factor of two, which method misses more often, and does the interval of (d) overlap the 500-day historical VaR? Model: B1.

B4: Cornish–Fisher vs Empirical Quantiles [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ Daily log returns in % (data/market): S&P 500 and gold 2000–2026, BET 2000–2026, Bitcoin 2014–2026, EUR/RON (BNR reference rate) 2005–2026; full samples.

□ Tasks

- ▶ (a) Compute the sample skewness S and excess kurtosis K of the returns of each series.
- ▶ (b) Compute VaR 1% and VaR 2.5% three ways: Normal, Cornish–Fisher (formula of A6) and historical.
- ▶ (c) Build a 95% percentile bootstrap interval (1 000 i.i.d. resamples) for the historical VaR 1% and check whether the Cornish–Fisher value lies inside.

□ Report

- ▶ One table with all numbers. Interpretation: is Cornish–Fisher inside the interval for any series, why does it work better at 2.5% than at 1%, and would moments from the last year help? Model: A1, A6, B1.

B5: Component VaR of a Four-Asset Portfolio [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ SPY, TLT, GLD (adjusted closes) and Bitcoin, data/market, 18 September 2014 – 18 September 2026; prices joined on the 3017 common days first, then daily simple returns in %.
- ▶ Equal weights of 25%, rebalanced daily; $\alpha = 1\%$ for VaR and $\alpha = 2.5\%$ for ES.

□ Tasks

- ▶ (a) Compute the Normal VaR 1% $z_{1-\alpha} \sqrt{w^\top \Sigma w}$, the marginal VaR and the component VaR of each asset [Tasche \(2008\)](#).
- ▶ (b) Compute the historical ES 2.5% of the portfolio and the contributions $\mathbb{E}[-w_i R_i \mid L_p \geq \text{VaR}_{2.5\%}]$ as averages over the tail days.
- ▶ (c) Compute the undiversified VaR $\sum_i z_{1-\alpha} w_i \sigma_i$.

□ Report

- ▶ A table of contributions and shares. Interpretation: is an equally weighted portfolio equally risky in each asset, how does it compare with the 40/30/20/10 portfolio of the lecture, and which weights would equalise the contributions? Model: A5.

B6: How Precise Is an ES Estimate? [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ S&P 500 (data/market/GSPC.INDX.csv) and BET, daily log returns in %, 2000–2026; two samples each: full and last 500 days.

□ Tasks

- ▶ (a) Compute historical ES 2.5% (lecture convention), the tail count $n\alpha$ and the number k of losses the tail mean actually averages.
- ▶ (b) Build a 95% percentile interval with the i.i.d. bootstrap [Efron \(1979\)](#), 2 000 resamples.
- ▶ (c) Repeat (b) with the moving-block bootstrap, blocks of 20 days [Künsch \(1989\)](#).

□ Report

- ▶ A table with $n\alpha$, ES and both intervals. Interpretation: why is the block interval wider on the full sample, and does a 500-day unconditional ES estimate pin ES down to within 0.3 percentage points [Pele & Mazurencu-Marinescu-Pele \(2026\)](#)? Model: B1.

B7: CAViaR on BET [Proposed]: Task

Solution discussed in class

□ Data and assumptions

- ▶ BET index (Bucharest Stock Exchange), data/market/BET.csv, daily closes 6 January 2000 – 18 September 2026; daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$, loss $L = -r$; weekdays only (the file has no holiday records).
- ▶ CAViaR with symmetric absolute value: $\text{VaR}_t = \beta_0 + \beta_1 \text{VaR}_{t-1} + \beta_2 |r_{t-1}|$, $\alpha = 1\%$, $\text{VaR}_1 = -\hat{q}_\alpha$ of the first 300 returns (their upper loss quantile), these days kept in the estimation sample [Engle & Manganeli \(2004\)](#).

□ Tasks

- ▶ (a) Estimate β on 2000–2015 by minimising the pinball loss $\sum_t (\alpha - 1\{r_t < -\text{VaR}_t\})(r_t + \text{VaR}_t)$: 10^4 random starts in $(0, 1)^3$, the best 10 refined by simplex and quasi-Newton steps.
- ▶ (b) With β fixed, forecast VaR for 2016–2026 and compute the exceedance rate and the mean pinball loss; compare with FHS (GARCH re-estimated each January) and 500-day historical simulation.
- ▶ (c) For 20 February – 30 April 2020 count the exceedances and report the peak VaR of each method.

□ Report

- ▶ The estimates, a comparison table and the COVID-19 numbers. Interpretation: what does $\hat{\beta}_1$ say about the memory of BET risk, how does CAViaR react to the crash compared with FHS, and are the exceedance counts different? (formal tests in Chapter 8). Model: B3; lecture, CAViaR on the S&P 500.

C1: ES Capital, BVB vs S&P 500 [Open]: Task

□ Data and assumptions

- ▶ BVB basket: Banca Transilvania, OMV Petrom, BRD, Transgaz, Romgaz, Nuclearelectrica, Electrica, Transelectrica (data/market, adjusted closes), equal weights, rebalanced daily; S&P 500 portfolio: SPY.
- ▶ Last ten years (from 2016-09-20), daily simple returns R_t in % in local currency, 1 million invested; a simplified illustration inspired by FRTB [Basel Committee \(2019\)](#), not the full capital calculation of [Basel Committee, MAR33](#).

□ Tasks

- ▶ (a) Compute the one-day historical ES 2.5% of each portfolio (loss $-R_t$).
- ▶ (b) Compute the 10-day ES 2.5% three ways: $\sqrt{10}$ scaling; overlapping compounded losses $100 [1 - \prod_{i=1}^{10} (1 + R_{t+i}/100)]$; filtered Monte Carlo from today's AR(1)-GARCH(1,1) state on the log returns, each path converted to a simple loss.
- ▶ (c) Stress window: the 250 consecutive days with the largest one-day historical ES 2.5%; compute the stressed ES inside it at 1 day and, by $\sqrt{10}$ scaling, at 10 days.
- ▶ (d) Capital per million = $m_c \times$ stressed ES, $m_c = 1.5$, for a liquidity horizon of 10 days (large cap) and of 20 days (small cap, a further factor $\sqrt{20/10}$).
- ▶ (e) 95% moving-block bootstrap intervals (20-day blocks, 2000 resamples) for (a) and for the one-day stressed ES of (c), with the stress window held fixed.

□ Report

- ▶ A table with all numbers and the intervals of (e), and a one-page argument: which portfolio needs more capital, and how do thin trading and currency (RON vs USD) change the answer?

C2: Critique an AI Answer [Proposed]

Solution discussed in class

The prompt sent to an artificial intelligence (AI) assistant

- “px holds daily BET closes, 2000–2026. Give me the one-day historical VaR 1% and ES 2.5% as positive losses in %, and the 10-day VaR 1%.”

The assistant's answer

```
import numpy as np
r = 100 * np.log(px).diff().dropna()           # daily log returns, %
loss = -r                                       # loss in % of the position
var1 = np.percentile(r, 99)                   # VaR 1%
es25 = loss[loss >= np.percentile(loss, 99)].mean() # ES 2.5%
var10 = 10 * var1                             # 10-day VaR 1%
```

- “VaR 1% = 3.95%, ES 2.5% = 6.28%, 10-day VaR 1% = 39.46% of the position.”

C2: Critique an AI Answer [Proposed]: Task

Solution discussed in class

▣ Data and assumptions

- ▶ The same BET data as in B1 (2000–2026, daily log returns in %); the code on the previous slide runs without errors, but the answer contains **three** mistakes typical for VaR and ES; everything else is correct.

▣ Tasks

- ▶ (a) For each mistake, name the line and explain in one sentence why it is wrong, using the definitions $\text{VaR}_\alpha = -q_\alpha(r)$ and $\text{ES}_\alpha = \text{mean loss beyond VaR}_\alpha$.
- ▶ (b) Say how you would detect it: a count of observations, a number from B1, or a slide of the lecture.
- ▶ (c) Write the corrected line(s) and recompute the three numbers.
- ▶ Hint: count how many daily losses es25 averages, and how many it should average.

▣ Report

- ▶ Log each mistake as in `AI_ERRORS.md`: what was asked, what the AI answered, what was wrong, how you found out, and the correct version. Model: B1.

C3: Deep Quantile Regression for VaR, a Project Seed [Open]: Task (1/2)

▣ Context and data

- ▶ Lecture 7, case study: [Chronopoulos, Raftapostolos & Kapetanios \(2024\)](#) replace the linear map of quantile regression by a feed-forward network $G_\alpha(x_t, w)$ with ReLU hidden layers, fitted by minimising the pinball loss plus a penalty (Eq. 10); on the S&P 500, with the last 2 000 days to August 2020 out of sample, all penalised deep networks had an RMSFE well below that of linear quantile regression at $\alpha = 1\%$ (Table 6).
- ▶ Daily log returns in %, $r_t = 100 (\ln P_t - \ln P_{t-1})$, from data/market: S&P 500 from 2 January 1990 to 31 August 2020 (the paper starts in September 1985), and the BET for the extension, each on its own calendar.

▣ Tasks

- ▶ (a) Keep the last 2 000 returns as the evaluation sample and the earlier ones as the tuning sample; on the tuning sample only, build the predictors of Eqs. 14 and 16: the GARCH(1,1) volatility σ_t estimated by QML, and the SAV CAViaR pair $(\text{VaR}_{t-1}, |r_{t-1}|)$ from a CAViaR fitted by the pinball loss [Engle & Manganeli \(2004\)](#).
- ▶ (b) At $\alpha = 1\%$ and 5% , fit a linear quantile regression (the benchmark) and a deep quantile network with ReLU hidden layers, a linear output and a ridge penalty; choose depth and width in $\{1, 5, 10\}$, the learning rate and λ from five values in $[0.001, 0.01]$ and the batch size in $\{10, 20\}$ by the time-series CV of Eq. 12 ($\delta = 5$, $\kappa = 20$), with the Adam optimiser and early stopping.

C3: Deep Quantile Regression for VaR, a Project Seed [Open]: Task (2/2)

▣ Tasks (continued)

- ▶ (c) With the parameters fixed, forecast $\widehat{\text{VaR}}_{t+10|\mathcal{F}_t} = G_\alpha(x_t, w^*)$ as in Eq. 23 for every day of the evaluation sample; compute the RMSFE of each network relative to the linear benchmark and the DM test [Diebold & Mariano \(1995\)](#) on the pinball losses with the correction of [Harvey, Leybourne & Newbold \(1997\)](#), and set your numbers next to the GARCH and SAV rows of Table 6.
- ▶ (d) Extension: re-estimate both models once a year on a rolling window and forecast September 2020 – 2026 for the S&P 500 and the BET; as an open step, train the same network with two outputs (v_t, e_t) on the FZ0 loss of [Patton, Ziegel & Chen \(2019\)](#) to forecast VaR and ES jointly.

▣ Report

- ▶ A table per index, predictor set, level and period: the selected architecture, the relative RMSFE, the DM statistic, the mean pinball loss, the exceedance rate and the number of exceedances.
- ▶ Interpretation (one page): does the deep network keep its advantage over linear quantile regression with a shorter tuning sample, after 2020 and on the BET, and what does the size of the gain say about how non-linear VaR is in its predictors?

Common Pitfalls

- ▣ Mixing sign conventions: a VaR reported as a negative return in one table and a positive loss in another
- ▣ Using the Normal distribution or Cornish–Fisher with daily kurtosis of 10
- ▣ Reporting ES 1% from 500 days without saying it averages 5 numbers
- ▣ Scaling by $\sqrt{10}$ a series with positive autocorrelation (BET, the BVB basket)
- ▣ Reading money weights as risk weights: a 10% crypto position can carry over a third of the risk
- ▣ Log returns summed across assets: portfolio losses need simple returns

Take-Home Assignment (Team Project)

- Choose a portfolio of at least three assets not covered in class (BVB stocks, a currency, a crypto-asset)
- Estimate one-day $\text{VaR}_{1\%}$ and $\text{ES}_{2.5\%}$ with historical, filtered and extreme-value methods, each with a bootstrap interval
- Split the ES into Euler contributions; from equal weights (long-only, fully invested, historical ES 2.5%, sample-mean expected return), minimise ES subject to the same expected return and aim for a 20% reduction; if it is infeasible, report the minimum attainable ES
- Log every AI error you catch in `AI_ERRORS.md` (at least three per project)
- Deliver a Quantlet folder (code, MetaInfo, charts) and three slides: numbers, interpretation, limitations

Further Study on Quantinar

Advanced courses on quantinar.com

- ▣  Measuring Statistical Risk
- ▣  XFG Advanced Methods in Quantitative Finance

Code and data

- ▣ Seminar notebook (Colab); Quantlets: github.com/danpele/MFM/tree/main/Quantlets/Ch_07

References (1/3)

- ▣ Acerbi, C., Tasche, D. (2002). [On the coherence of expected shortfall](#). *Journal of Banking & Finance*, 26(7), 1487–1503.
- ▣ Artzner, P., Delbaen, F., Eber, J.-M., Heath, D. (1999). [Coherent measures of risk](#). *Mathematical Finance*, 9(3), 203–228.
- ▣ Barone-Adesi, G., Giannopoulos, K., Vosper, L. (1999). [VaR without correlations for portfolios of derivative securities](#). *Journal of Futures Markets*, 19(5), 583–602.
- ▣ Basel Committee on Banking Supervision (2019). [Minimum capital requirements for market risk](#). Bank for International Settlements.
- ▣ Chen, S.X. (2008). [Nonparametric estimation of expected shortfall](#). *Journal of Financial Econometrics*, 6(1), 87–107.
- ▣ Christoffersen, P., Gonçalves, S. (2005). [Estimation risk in financial risk management](#). *Journal of Risk*, 7(3), 1–28.
- ▣ Chronopoulos, I., Raftapostolos, A., Kapetanios, G. (2024). [Forecasting Value-at-Risk using deep neural network quantile regression](#). *Journal of Financial Econometrics*, 22(3), 636–669.
- ▣ Cornish, E.A., Fisher, R.A. (1938). [Moments and cumulants in the specification of distributions](#). *Revue de l'Institut International de Statistique*, 5(4), 307–320.
- ▣ Diebold, F.X., Mariano, R.S. (1995). [Comparing predictive accuracy](#). *Journal of Business & Economic Statistics*, 13(3), 253–263.

References (2/3)

- ▣ Efron, B. (1979). [Bootstrap methods: another look at the jackknife](#). *Annals of Statistics*, 7(1), 1–26.
- ▣ Engle, R.F., Manganelli, S. (2004). [CAViaR: conditional autoregressive value at risk by regression quantiles](#). *Journal of Business & Economic Statistics*, 22(4), 367–381.
- ▣ Ferro, C.A.T., Segers, J. (2003). [Inference for clusters of extreme values](#). *Journal of the Royal Statistical Society: Series B*, 65(2), 545–556.
- ▣ Franke, J., Härdle, W.K., Hafner, C.M. (2019). [Statistics of Financial Markets: An Introduction](#), 5th ed. Springer.
- ▣ Harvey, D., Leybourne, S., Newbold, P. (1997). [Testing the equality of prediction mean squared errors](#). *International Journal of Forecasting*, 13(2), 281–291.
- ▣ Künsch, H.R. (1989). [The jackknife and the bootstrap for general stationary observations](#). *Annals of Statistics*, 17(3), 1217–1241.
- ▣ McNeil, A.J., Frey, R. (2000). [Estimation of tail-related risk measures for heteroscedastic financial time series: an extreme value approach](#). *Journal of Empirical Finance*, 7(3–4), 271–300.
- ▣ McNeil, A.J., Frey, R., Embrechts, P. (2015). [Quantitative Risk Management: Concepts, Techniques and Tools](#), revised ed. Princeton University Press.
- ▣ Patton, A.J., Ziegel, J.F., Chen, R. (2019). [Dynamic semiparametric models for expected shortfall \(and Value-at-Risk\)](#). *Journal of Econometrics*, 211(2), 388–413.

References (3/3)

- ▣ Pele, D.T., Mazurencu-Marinescu-Pele, M. (2026). Finite-sample precision limits for expected shortfall forecast comparisons. *Mathematics*, 14(13), 2316.
- ▣ Rockafellar, R.T., Uryasev, S. (2000). Optimization of conditional value-at-risk. *Journal of Risk*, 2(3), 21–41.
- ▣ Tasche, D. (2008). Capital allocation to business units and sub-portfolios: the Euler principle. arXiv:0708.2542.